WEEK 2 ASSIGNMENT 2

Q1. A bank wants to predict whether a loan application will be approved or rejected.

- **Is this a classification or regression problem?**
- **Which algorithm among Decision Tree, KNN, and Linear Regression would be most suitable?**

A1-

- Type of Problem: This is a classification problem because the outcome you want to predict falls into distinct, categorical buckets: Approved or Rejected.
- Most Suitable Algorithm: A Decision Tree (or KNN) would be most suitable. Linear Regression is not appropriate because it is meant for predicting continuous numbers (like price or salary), not yes/no categories.

Q2. Explain the difference between a feature and a label using a real-world example.

A2-

- Feature: The inputs or "clues" used to make a prediction.
- Label: The final outcome, answer, or "target" you are trying to predict.
- Real-World Example (House Pricing): * Features: The number of bedrooms, the square footage, and the neighborhood.
- Label: The final selling price of the house.

Q3. A Decision Tree predicts that a customer will purchase a product. Explain how the model reaches this prediction.

A3-

A Decision Tree reaches a prediction by acting like a structural flowchart or a game of "20 Questions."

- It starts at the very top question (the Root Node), checking an initial condition like: Is the customer's annual income > $50,000?
- Based on the answer (Yes or No), it follows a specific Branch down to the next criteria question (e.g., Has this customer browsed the item for more than 10 minutes?).
- It repeats this step-by-step narrowing process until it lands on a terminal bucket (the Leaf Node), which provides the final decision: Will Purchase.

Q4. Why is KNN considered a supervised learning algorithm? Can KNN perform classification if labels are not available?

A4-

- Why it is Supervised: KNN is a supervised algorithm because it relies entirely on a historical dataset that already contains the correct answers (labels) to learn how to classify new points.
- Can it work without labels? No. If labels are missing, KNN cannot perform classification. It needs those reference labels because it determines a new point's identity solely by looking at the known classes of its closest neighbors.

Q5. Suppose a KNN model uses: K = 5 and the nearest neighbors are: Yes, Yes, No, Yes, No What will be the prediction and why?

A5-

- The Prediction: Yes
- Why: KNN makes its final choice based on a simple democratic majority vote among the closest neighbors. Out of the 5 nearest neighbors, 3 voted "Yes" and 2 voted "No". Since "Yes" has the majority, the model assigns "Yes" as the final prediction.

Q6.

Explain the equation of Linear Regression:

$$y = mx + b$$

$$m$$

$$b$$

What do x, y, m, and b represent?

A6-

This equation represents a straight trend line drawn through your data points:

- y (The Dependent Variable): The final answer or value you are trying to predict (e.g., Monthly Spend).
- x (The Independent Variable): The input feature you are using to make the prediction (e.g., Watch Hours).
- m (The Slope / Coefficient): The weight or steepness of the line. It tells you how much y changes every time x increases by one unit.
- b (The Y-Intercept): The starting baseline value of y when the feature x is exactly zero.

Q7. What is correlation? $y = mx + b$ What does each of the following values indicate? +1 0 -1

A7-

- +1 (Perfect Positive Correlation): As one variable goes up, the other variable goes up by a perfectly proportional amount (e.g., more watch hours = higher spend).
- 0 (No Correlation): The two variables have absolutely no linear relationship to each other (e.g., a user's shoe size has zero impact on their streaming subscription choices).
- -1 (Perfect Negative Correlation): As one variable goes up, the other goes down in a perfect straight-line relationship (e.g., as user age increases, maybe overall watch hours decrease).

Q8. A disease detection model produced the following result: Actual Condition Prediction Disease Present Healthy. What type of prediction is this? Why can this be dangerous?

A7-

- Type of Prediction Error: This is a False Negative error (the model predicted the patient was healthy, but they actually carry the disease).
- Why it is Dangerous: In medical diagnostics, a False Negative is highly dangerous because a sick patient is sent home under the false impression that they are perfectly fine. This delays vital medical treatment, allows the underlying condition to worsen, and could potentially lead to life-threatening consequences.

Q9. What is the purpose of a Confusion Matrix? How is it different from a Correlation Matrix?

A9-

- Confusion Matrix: A grid layout used to evaluate the operational performance of a classification model. It directly contrasts the model's categorical guesses against the true labels by breaking down the exact counts of True Positives, True Negatives, False Positives, and False Negatives.
- Correlation Matrix: A statistical grid table used during data exploration to show the strength of relationships between features. It simply displays correlation scores ranging from -1 to 1 to see how tightly two columns move together.

Q10. Compare the following algorithms in one or two lines each: Decision Tree KNN SVM Linear Regression Neural Network Focus on the main idea behind each algorithm.

A10-

- Decision Tree: Predicts outcomes by passing data down a sequential flowchart of simple yes/no questions until it hits a final decision leaf.
- KNN (K-Nearest Neighbors): Predicts an unknown data point's class by identifying its closest historical neighbors and taking a simple majority vote.
- SVM (Support Vector Machine): Finds the optimal geometric boundary line (or hyperplane) that creates the widest possible gap to cleanly separate different classes.
- Linear Regression: Draws a single best-fitting straight trend line (y=mx+b) through continuous data points to predict numerical values.
- Neural Network: Mimics the interconnected structure of the human brain using layers of interconnected nodes to discover complex, non-linear patterns in large datasets.